Trigonometric functions on the unit circle

1

a $\frac{1}{2}$ b $\frac{\sqrt{3}}{2}$ c $\frac{\sqrt{2}}{2}$ d $\frac{\sqrt{3}}{2}$

e 0 f 1 g 1 h $\frac{1}{\sqrt{3}}$

2

a $-\frac{1}{2}$ b $-\frac{1}{2}$ c -1 d $-\frac{1}{2}$

e $\frac{\sqrt{3}}{2}$ f -1 g $-\frac{1}{\sqrt{3}}$ h $-\frac{1}{2}$

3

a $\frac{1}{y}$ b $\frac{1}{x}$ c $\frac{x}{y}$

4

a $\frac{2}{\sqrt{3}}$ b 1 c $-\frac{2}{\sqrt{3}}$ d 2

e $-\frac{\sqrt{2}}{-\sqrt{2}}$ f $-\sqrt{2}$ g $-\sqrt{2}$ h 1

5

a $\frac{2}{\sqrt{3}}$ b $\frac{2}{\sqrt{3}}$ c -2 d $-\frac{2\sqrt{3}}{3}$

e $\sqrt{3}$ f $\sqrt{2}$ g $-\frac{2\sqrt{3}}{3}$ h $\sqrt{3}$

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a

i Quadrants 1 and 2 ii Quadrants 3 and 4

b

i Quadrants 1 and 4 ii Quadrants 2 and 3

c

i Quadrants 1 and 3 ii Quadrants 2 and 4

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a

i $\frac{\sqrt{3}}{2}$ ii $\frac{1}{2}$ iii $\sqrt{3}$ iv $\frac{2}{\sqrt{3}}$

v 2 vi $\frac{1}{\sqrt{3}}$

b

i $\frac{\sqrt{21}}{5}$ ii $-\frac{2}{5}$ iii $-\frac{\sqrt{21}}{2}$ iv $\frac{5}{\sqrt{21}}$

v $-\frac{5}{2}$ vi $-\frac{2}{\sqrt{21}}$

c

i $\frac{-1}{\sqrt{2}}$ ii $\frac{1}{\sqrt{2}}$ iii -1 iv $-\sqrt{2}$

v $\sqrt{2}$ vi -1

8

a

i $\frac{1}{\sqrt{2}}$ ii $\frac{1}{\sqrt{2}}$ iii 1

b

i 135° ii 225° iii 315°

c

i $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ii $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ iii $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d

i $\sin 45^\circ$ ii $-\cos 45^\circ$ iii $-\tan 45^\circ$ iv $-\sin 45^\circ$

e

i $\frac{1}{\sqrt{2}}$ ii $-\frac{1}{\sqrt{2}}$ iii -1 iv $-\frac{1}{\sqrt{2}}$

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a No solution provided.

b No solution provided.

c No solution provided.

d No solution provided.

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